

Who Wrote It? Identifying LLMs from Their Responses

(CMPSC 448; modified 9/17/2026)

Deadline: 11:59pm on Oct. 11, 2026

Project Overview: As Large Language Models (LLMs) such as GPT, Claude, Gemini, Llama, and Qwen increasingly generate the text we read online, we face an emerging digital authorship crisis: Can we still tell who—or what—wrote a piece of text? Different LLMs may leave distinct linguistic, stylistic, and structural “fingerprints” in their responses, raising an intriguing question: [Can machine learning identify the AI behind the text?](#)

In this project, each team can **obtain** or **construct** data with at least three components: “(LLM_name, LLM_input, LLM_output)”. Your dataset should contain responses from at least 3 different LLM families. Examples include, but are not limited to: GPT, Claude, Gemini, Llama, Qwen, DeepSeek, other LLMs. You may obtain data from: publicly released datasets; publicly available LLM outputs; LLM APIs or interfaces; interactions that you generate yourself; your own LLM interactions that you are comfortable using for this project. You should clearly document where your data came from and how it was collected or constructed. Do not include private, sensitive, confidential, copyrighted-inappropriately-reproduced, or third-party information that you do not have the right to use. Try to keep the number of examples from different LLMs reasonably balanced.

Teams: [You may work individually or in a team of up to 5 students.](#) Each team should choose a team leader who is responsible for:

- leading and coordinating the project;
- submitting the project;

Required Models: You must implement and evaluate the following two models: [Convolutional Neural Network \(CNN\) and Recurrent Neural Network \(RNN, like LSTM\)](#). You may additionally experiment with other classifiers or combine models to obtain better performance. However, CNN and RNN are required.

Research Questions (each team must investigate **at least RQ1 and RQ2.** **You are encouraged to further investigate RQ3 and/or RQ4 for extra credit. Each additional RQ is worth 5 extra-credit points.**).

- RQ1 — Can we identify which LLM generated a response?
Use LLM_output as input, conduct classification to detect which LLM generated it. Report the performance of your CNN&RNN performance
- RQ2 — Does the user's prompt help identify the LLM?
Compare compare classifiers using: Input only; Output only; Input + Output
Questions you may consider: i) Can the LLM be predicted from the user's prompt even before seeing its response? ii) Does including the input improve classification? iii) If input-only classification works surprisingly well, why might this happen?
- RQ3 — Do LLM fingerprints generalize across tasks or domains? **(Extra credit "5")**
Investigate whether a classifier trained on one type of task can identify LLMs on another type of task.
For example: Training on problems like coding + mathematics + factual questions, but Testing on writing. Or Training on general question answering but Testing on coding.
Questions you may consider: i) Does performance decrease on unseen tasks? ii) Are some models more consistently identifiable across domains? iii) Is the classifier learning a general LLM fingerprint or task-specific patterns?
- RQ4 — What characteristics distinguish different LLMs? **(Extra credit "5")**
Classification accuracy tells us whether models can be distinguished. This RQ asks why. Analyze linguistic or structural characteristics such as: response length; vocabulary; lexical diversity; formatting, etc.

Questions you may consider: i) What signals does the classifier appear to use? ii) Are the signals linguistic, structural, or both? iii) Does removing certain signals substantially reduce classification performance?

What You Can Use

- PyTorch; NumPy; pandas; scikit-learn; NLTK; spaCy; other standard Python/ML packages;
- publicly available datasets.

- You may use LLMs such as ChatGPT, Claude, Gemini, etc. to help you: understand concepts; debug code; brainstorm experiments; interpret results. However, you are responsible for understanding, verifying, and explaining everything submitted by your team.

What You Need to Submit

- Submit the URL of your GitHub repository. Your repository should include:
 - Code
 - Project Report in pdf (names of team leader and members included)

Evaluation:

- **PDF quality (80%):** we score by the following five aspects:
 - Problem definition and Dataset curation (20 points)
 - Your CNN/RNN implementation and how you train it (i.e., your model details and training details) (20 points)
 - The results and how they are presented (20 points)
 - Any in-depth analyses/experiments (20 points)
 - Lessons&Experience you learned in this project (20 points)

For individual participants, points will be deducted in units of 5; for group participants, points will be deducted in units of $4+2 \cdot G$ where “G” refers to group size (2,3,4 or 5).

- **(optional) Presentation (20%):** Each participant (individual or group) can choose whether you want to present or not. You can get a default score **17** (out of max 20) for the presentation part if you choose NOT (such as your scores of other parts can secure an “A” for you). If you choose to present, the evaluation criteria include: slide quality, work quality, presentation skills, question answering, etc.) For individual presentations, points will be deducted in units of 2; for group presentations, points will be deducted in units of 4.
- If it is a group project, each team member gets the same score except the team leader has an extra credit score “**3**”.